

Smart Waste Management System

For Metropolitan Cities

Final Year Project Documentation

SmartInternz Externship Program — 2026

Submitted By: Prithviraj Prashant Nimbalkar

Project Domain: IoT · Smart Cities · Web Development · AI

1. Project Overview

The Smart Waste Management System (SWMS) is an advanced full-stack web application that simulates a real-world IoT-powered waste monitoring platform for metropolitan cities. It integrates real-time bin fill-level tracking, AI-based route optimization, live analytics dashboards, and a REST API for IoT device connectivity.

The system monitors 12 smart bins distributed across 6 zones of Pune Metropolitan Region, providing municipal authorities with actionable insights to optimize collection schedules, reduce fuel costs, and maintain city hygiene standards.

Built using Python Flask, Leaflet.js maps, Chart.js analytics, and a professional dark-sidebar UI, this project demonstrates a complete end-to-end smart city solution aligned with UN Sustainable Development Goals 11, 12, 14, and 15.

2. Objectives

- Design and develop a real-time IoT bin monitoring system using simulated ultrasonic sensor data
- Implement an AI-powered route optimization algorithm (Greedy Nearest-Neighbour with Haversine distance)
- Build a professional analytics dashboard with live charts, KPI cards, and zone-wise analysis
- Create an interactive city map with color-coded bin markers and animated truck route visualization
- Develop a complete REST API for IoT device integration and mobile app connectivity
- Align the solution with UN SDG 11 (Sustainable Cities) and SDG 12 (Responsible Consumption)

3. Technologies Used

Layer	Technology	Purpose
Backend	Python 3 + Flask 3.1	Web server, routing, REST API
Frontend	HTML5 + Custom CSS3	Professional dark-sidebar UI
Maps	Leaflet.js + OpenStreetMap	Interactive city map with bin markers
Charts	Chart.js 4.x	Bar, doughnut, pie, line charts
Algorithm	Greedy Nearest-Neighbour	Optimal truck route planning
Distance	Haversine Formula	GPS-accurate distance calculation
IoT Sim	Time-seeded RNG	Realistic fill-level simulation
API	Flask REST (JSON)	IoT device & mobile integration
Version Control	Git & GitHub	Source code management

4. Folder Structure

```
Smart-Waste-Management-System/
├── Documentation/
│   └── Project_Documentation.docx
├── static/css/
│   └── style.css
├── templates/
│   ├── _sidebar.html      ← Reusable sidebar component
│   ├── home.html         ← Landing page
│   ├── dashboard.html    ← Live analytics dashboard
│   ├── map.html          ← Interactive map + route
│   ├── bins.html         ← Bin monitoring grid
│   └── reports.html      ← Weekly KPI reports
├── app.py                ← Main Flask application
├── requirements.txt
└── README.md
```

5. System Architecture

5.1 Overall Flow

- IoT sensors (simulated) report bin fill levels every 15 minutes
- Flask backend processes data, computes route, generates alerts

- Jinja2 templates render live data to the browser
- JavaScript (Chart.js, Leaflet) renders interactive visuals client-side
- REST API endpoints allow IoT devices and mobile apps to push/pull data

5.2 Bin Network — 12 Smart Bins across 6 Zones

Bin ID	Location	Zone	Waste Type	Capacity
BIN001	MG Road Junction	Zone A	General	100 L
BIN002	FC Road Market	Zone A	Recyclable	100 L
BIN003	Shivaji Nagar Station	Zone B	Organic	150 L
BIN004	Koregaon Park Gate	Zone B	General	100 L
BIN005	Baner Tech Hub	Zone C	Recyclable	200 L
BIN006	Hinjewadi IT Park	Zone C	General	200 L
BIN007	Wakad Market	Zone C	Organic	100 L
BIN008	Kothrud Square	Zone D	General	150 L
BIN009	Sinhagad Road	Zone D	Recyclable	100 L
BIN010	Hadapsar Industrial	Zone E	Hazardous	200 L
BIN011	Viman Nagar Airport Rd	Zone E	General	100 L
BIN012	Aundh Central Park	Zone F	Organic	150 L

6. Key Features

6.1 Live Dashboard

- 4 KPI stat cards: Total Bins, Critical Bins (≥85%), Warning Bins (60–84%), Average City Fill %
- Chart.js grouped bar chart: weekly waste collected vs recycled (kg)
- Doughnut chart: bin status distribution (OK / Warning / Critical)
- Real-time alert feed with severity icons and time labels
- Zone-wise average fill percentage grid (6 zones)
- Critical bins table with immediate collection action button

6.2 Interactive Map

- Leaflet.js map centered on Pune Metropolitan Region
- 12 color-coded bin markers: Red (Critical), Yellow (Warning), Green (OK)
- Popup cards for each bin showing fill level, type, and predicted full time
- AI-computed truck route shown as animated dashed polyline
- Numbered stop markers (1, 2, 3...) on the optimal route
- Route stats: stops count, total distance (km), estimated travel time

6.3 Bin Monitor

- Responsive card grid — all 12 bins with fill bars, status badges
- Filter by Zone (A–F) and Waste Type (General, Recyclable, Organic, Hazardous)
- Predicted full time based on fill rate heuristic
- Color-coded top border per card (Green/Yellow/Red by status)

6.4 Reports & KPIs

- Weekly collection trend bar chart (collected vs recycled)
- Waste type distribution pie chart
- Fill level trend line chart (7-day history)
- KPI cards: total collected, recycled, recycling rate %, active bins
- Complete bin status table with fill progress bars

7. Route Optimization Algorithm

7.1 Greedy Nearest-Neighbour

The system uses a Greedy Nearest-Neighbour algorithm to compute the optimal garbage truck collection route. Only bins with Critical status ($\geq 85\%$ full) are included in the route to minimize unnecessary travel.

7.2 Haversine Distance Formula

GPS coordinates of bins are used to compute accurate real-world distances using the Haversine formula, which accounts for the Earth's curvature:

$$d = 2R \times \arcsin(\sqrt{\sin^2(\Delta\text{lat}/2) + \cos(\text{lat}1) \cdot \cos(\text{lat}2) \cdot \sin^2(\Delta\text{lng}/2)})$$

This gives distances in kilometers, enabling realistic route time estimation at 30 km/h average truck speed.

8. IoT Sensor Simulation

Each bin's fill level is simulated using a time-seeded random generator that mirrors realistic ultrasonic sensor behavior:

- Fill levels are consistent within each hour (same as a real sensor reading every 15 min)
- Fill levels change across hours (simulates waste accumulation over time)
- Each bin has a unique seed based on its ID to ensure different realistic readings
- Alert status is determined by thresholds: OK ($< 60\%$), Warning ($60\text{--}84\%$), Critical ($\geq 85\%$)

9. REST API Reference

Endpoint	Method	Description	Response
/api/bins	GET	All bin status with fill levels	JSON array
/api/bins/<id>	GET	Single bin real-time data	JSON object
/api/bins/<id>/update	POST	IoT device pushes fill level	Action decision
/api/route	GET	AI-optimized truck route	Ordered bin list
/api/stats	GET	City-level aggregated stats	Summary JSON

9.1 Sample API Response — /api/stats

```
{"total_bins": 12, "critical": 4, "warning": 3, "ok": 5, "avg_fill": 61.5}
```

10. Setup & Run Instructions

Step 1: Install Dependencies

```
pip install -r requirements.txt
```

Step 2: Run Flask Application

```
python app.py
```

Step 3: Open in Browser

```
http://localhost:5000
```

11. SDG Alignment

SDG Goal	Title	Contribution
SDG 11	Sustainable Cities	Smart bin monitoring reduces urban waste overflow
SDG 12	Responsible Consumption	Recycling rate tracking promotes responsible disposal
SDG 14	Life Below Water	Prevents landfill runoff from reaching waterways
SDG 15	Life on Land	Reduces plastic waste and land pollution

12. Conclusion

The Smart Waste Management System successfully demonstrates a complete, industry-level IoT and AI web platform for metropolitan city waste management. By combining real-time bin monitoring, intelligent route optimization, professional analytics, and a developer-

friendly REST API, the system delivers genuine value for municipal waste management authorities.

The platform is immediately deployable and can be extended to integrate real ultrasonic sensors (Arduino/Raspberry Pi via MQTT), SMS/email alert systems, and machine learning models for predictive waste forecasting. It represents a practical application of modern web technologies to solve a critical urban sustainability challenge.

Github Link: [<https://github.com/Pruthviraj-01/Smart-Waste-Management-System-Main.git>]

Demo Link:

[https://drive.google.com/file/d/1OwLa97n2w0KFCPWFcfSKB_4p0ZOPHL4q/view?usp=sharing]

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